Manuel Caipo

Machine Learning Engineer | AI Researcher



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Machine Learning Engineer combining a mechanical engineering foundation with graduate training in numerical methods and machine learning theory, focused on physics-aware and reliable AI for complex cyber-physical systems. Strong ability to interpret sensor behavior, analyze physical degradation mechanisms, and translate them into physically consistent models for anomaly detection and remaining useful life prediction, deployed end-to-end on high-frequency time-series data with demonstrated industrial impact. My current research extends this foundation toward agentic AI, developing LLM-based agents that orchestrate specialized ML models across signals, logs, and engineering artifacts, together with knowledge distillation methods that turn historical cases and validated expert decisions into reusable and verifiable reasoning. The goal is explainable and reliable AI that accelerates engineering processes, reduces manual analysis effort, and keeps every decision traceable to its physical and data evidence.

PROFESSIONAL EXPERIENCE

06.2026 – Today

Mercedes-Benz AG, Sindelfingen, Germany
RD/EDD eDrive Software Integration – PhD Candidate

- Doctoral research, in cooperation with TUM's Chair of Sustainable Mobile Drivetrains, on context-aware multimodal learning for fault analysis in software-intensive eDrive systems.
- Design of agentic LLM approaches that orchestrate specialized ML fault classifiers across heterogeneous data sources (signals, logs, software and test artifacts).
- Focus on explainable and traceable diagnostics, distilling validated expert knowledge into verifiable diagnostic decisions to improve engineering processes.

11.2024 – 03.2026

Bosch Rexroth AG, Ulm, Germany
DC-SSD3 – Master's Thesis

- Master's thesis focusing on the development of an ML/AI-based condition monitoring module for *CytroConnect*, combining physical models, hydraulic domain knowledge, and data compliant with the *Fluid 4.0* standard.

DC-SSD3 – Data Scientist Working Student

- Built scalable ETL pipelines for high-frequency industrial time-series data (SQL/NoSQL) and integrated message brokers (Solace/MQTT).
- Developed and deployed ML/DL-based predictive models for anomaly detection and RUL prediction in hydraulic systems.
- Applied explainable AI methods to increase transparency, robustness, and acceptance of data-driven models.

04.2021 – 07.2023

Freeport-McMoRan Cerro Verde, Phoenix, U.S.A.
Data Analytics – Junior Data Scientist (1 → 2)

- Developed and deployed ML/DL models (LSTM, CNN, XGBoost) in Azure ML for anomaly detection, wear and RUL prediction of critical assets (FLSmidth primary crushers, Weir Minerals cyclone pumps), including SHAP/PDP explainability, continuous retraining and optimized SQL Stored Procedures for accelerated training data preparation.
- RUL-based proactive maintenance planning: +1.5% asset availability (≈$10M/year savings). Awarded Innova 2022, Showroom Award and President's Award.

Maintenance Reliability – Trainee → Junior Data Analyst

- Time-series analysis (ARIMA), RCA and automated KPI reporting (SQL, Dagster, Power BI) to shift maintenance from fixed intervals to condition-based strategies in SAP.
- Long-term performance and failure analysis of the concentrator plant: primary and secondary crushers, HPGR, ball mills, conveyor belts and cyclone pumps.

08.2020 – 03.2021

IMCO Servicios S.A.C, Arequipa, Peru
Engineering and Development – Junior Engineer

- Performed static and dynamic structural analyses with topology optimization in a BIM-oriented methodology, including architectural modeling (Tekla), FEM-based structural design (SAP2000, IdeaStatica, Inventor, Ansys Structural), and final plan review in AutoCAD 3D.

01.2020 – 06.2020

Freeport-McMoRan Cerro Verde, Arequipa, Peru
Maintenance Reliability – Intern

- Multiphysics CFD simulations (Ansys Fluent) to analyse wear mechanisms and evaluate protective coatings for service-life optimization.

PUBLICATIONS

2026 **Explainable Edge-AI Framework for Fault Diagnosis and Remaining Useful Life Prediction in Hydraulic Systems**. Published in *Discover Mechanical Engineering* (Springer Nature), 2026. DOI: 10.1007/s44245-026-00364-y [Z]

[Z] Article

TECHNICAL SKILLS

Programming & Scientific Computing: Python (NumPy, Pandas, SciPy, scikit-learn, PyTorch), SQL, C/C++, MATLAB; object-oriented design and numerical modeling

Machine Learning & AI Systems: End-to-end ML system design for predictive maintenance and RUL modeling; LSTM (Attention), CNN, XGBoost, Random Forest; time-series forecasting, anomaly detection; Explainable AI (SHAP, PDP)

Agentic AI & LLM Systems: LLM-based agents, multi-agent orchestration, tool calling; RAG, prompt engineering, LLM evaluation; knowledge distillation; LangChain, LangGraph, Hugging Face

Data Engineering & ETL: PostgreSQL, Snowflake, InfluxDB (SQL & NoSQL); Dagster, Apache Airflow; database schema design, batch and streaming pipelines

Cloud & Scalable ML Deployment: Azure ML (pipelines, model registry, deployment, automated retraining), Databricks; Kafka, MQTT; scalable cloud-based ML services

MLOps & DevOps: Git, Docker, CI/CD; experiment tracking, reproducible environments, containerized inference services

Industrial & Physical Systems: OPC UA integration, Siemens TIA Portal (PLC), Bosch Rexroth CtrlX Core; FEM (Ansys), CFD; physics-informed modeling for cyber-physical systems

EDUCATION

06.2026 – Today — **PhD Candidate – Chair of Sustainable Mobile Drivetrains**
Topic: Context-Aware Multimodal Learning for Fault Analysis in Software-Intensive Systems.
Technical University of Munich, Munich, Germany

10.2025 – 04.2026 [i] **Exchange Semester – Computational Science and Engineering** — 1,59
Focus: Numerical Optimization, High-Performance Computing.
Ulm University, Ulm, Germany

09.2024 – 04.2026 [i] **Master of Science – Advanced Precision Engineering** — 1,59
Furtwangen University of Applied Sciences, Villingen-Schwenningen, Germany
[ii] Thesis: Fluid 4.0-compliant framework for digital representation and graph-based neural modeling (published, Springer).

06.2021 – 02.2022 [i] **Postgraduate Diploma – Machine Learning & Deep Learning** — [Top 5 %] 19.6/20.0
Universidad Católica San Pablo, Arequipa, Peru

03.2015 – 12.2019 [i] **Bachelor – Mechanical Engineering** — [Top 1 %] 15.2/20.0
National University of San Agustín, Arequipa, Peru
[ii] *Official Recognition by the German State*
[iv] Thesis: Design of a 350 TPH copper concentrate conveyor belt with automatic hydraulic tensioning system and loading hopper

[i] Diploma

AWARDS & HONORS

2022 **Innova 2022 – Triple Award Winner** (1st Place Digital Transformation, Showroom & President's Award) – Freeport-McMoRan. *Machine & Deep Learning Life Data Analysis* with measurable impact (+1.5% asset availability, ≈>$10M USD/year). [Z]

2021 **Outstanding Bachelor's Thesis Award** – Engineering Sciences. [Z]

2019 **Top Graduate (Top 1%)** – Mechanical Engineering, National University of San Agustín, Peru. [Z]

[Z] Certificate

REFERENCES

2026 **Dr. Jörn Kretschmer, M.Sc.**, Hochschule Furtwangen University. Supervisor of my Master's thesis — Recommendation for PhD studies Joern.Kretschmer@hs-furtwangen.de. [Z]

2025 **Prof. Dr.-Ing. G. Ketterer**, Hochschule Furtwangen University, Dean of Studies M.Sc. *Advanced Precision Engineering*. Recommendation for PhD studies — gunter.ketterer@hs-furtwangen.de. [Z]

2023 **Travis Gaddie**, Freeport-McMoRan, Data Scientist III, Metals Optimization. Reference for career development and advanced studies — tgaddie@fmi.com. [Z]

2021 **Dr. L. Rodríguez B.**, National University of San Agustín (UNSA), Director of Mechanical Engineering & Bachelor's thesis supervisor. Recommendation for Master's studies — lrodriguez4@unsa.edu.pe. [Z]

[Z] Letter

SCHOLARSHIPS & GRANTS

2019 **Research Grant – SERESSA 2019**, Spain. 15th International School on Radiation Effects.

2017 **President of the Republic Scholarship – Peru**. Full scholarship awarded for academic excellence (Top 1%).

ADDITIONAL COURSES

2024 German Courses B2, C1 (Ulm vhs & Berlitz) [Z]
2022 Leadership and Soft Skills (UCSP) [Z]
2021 Hydraulic Systems (Bosch Rexroth) [Z]
2019 Pipeline Calculation (EADIC) [Z]
2020 Structural & Multiphysics Analysis with FEM (ESSS) [Z]
2019 Industrial Plant Systems (TECSUP) [Z]
2019 Autodesk Nastran – FEM & CFD (Autodesk) [Z]
2019 Autodesk Inventor – CAD (Autodesk) [Z]
2018 Process Improvement (Proavance) [Z]

[Z] Certificate